

A window has a frame percentage equal to f . What are the dimensions of the glass?

If a window has **overall dimensions** ($W \times H$) (width by height) and a **frame percentage** (f) (the fraction of the window area taken up by the frame), then the **glass area** is:

$$[A_{\text{glass}} = (1 - f)WH.]$$

But if you want the **actual glass dimensions** (glass width and glass height), you need to know **how the frame is distributed** around the edges.

If the frame is uniform (same thickness on all sides)

Let the frame thickness be t . Then:

$$[\text{Glass width} = W - 2t] [\text{Glass height} = H - 2t]$$

The glass area is:

$$[(W - 2t)(H - 2t) = (1 - f)WH.]$$

You can solve for t :

$$[(W - 2t)(H - 2t) = (1 - f)WH.]$$

Expanding:

$$[WH - 2t(W + H) + 4t^2 = (1 - f)WH,]$$

$$[-2t(W + H) + 4t^2 = -fWH,]$$

$$[4t^2 - 2(W+H)t + fWH = 0.]$$

This is a quadratic in t :

$$[t = \frac{(W+H) \pm \sqrt{(W+H)^2 - 4 \cdot 4 \cdot fWH}}{8}.]$$

Only the **smaller** root is physically meaningful.

If you want, tell me:

- the window width (W),
- height (H),
- frame fraction (f),

and I'll compute the exact glass width and height for you.